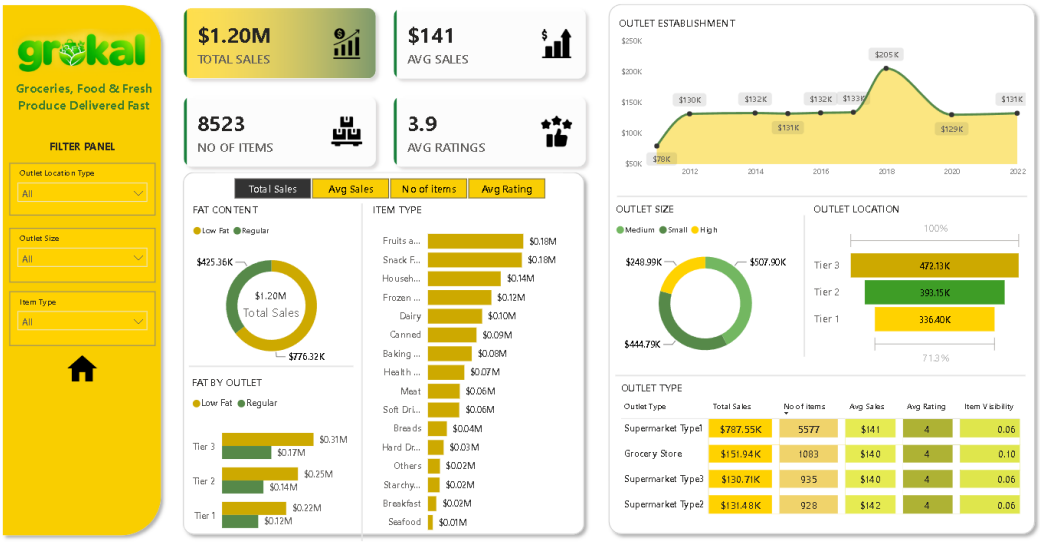
END-TO-END DATA ANALYTICS PROJECT

Grokal Sales & Outlet Performance Analysis

Python • SQL • Power Query • Power BI • DAX

An end-to-end retail analytics project focused on understanding sales performance, product-category contribution, and outlet-level patterns. The project is framed around the complete analytics lifecycle: data understanding, data cleaning, exploratory analysis, SQL-based investigation, KPI definition, visual analysis, and business reporting.



Power BI report visual used as the presentation layer of the broader analysis.

PROJECT SUMMARY

Business domain	Retail sales and outlet performance	**Analysis focus**	Sales, products, outlet segments, ratings
Data tools	Python and SQL for analysis and validation	**Reporting tool**	Power BI with DAX measures and slicers
Intended users	Business owners, sales managers, outlet teams	**Outcome**	Evidence-led comparisons and performance monitoring

Project goal: move from raw retail records to validated analysis and a clear business-facing report.

01 | Project Overview

Grokal Sales & Outlet Performance Analysis examines retail sales through product and outlet dimensions. The purpose is not only to present charts, but to follow an analytics workflow that checks data quality, explores patterns, validates calculations, and translates findings into information that business teams can use.

Business problem

Sales and outlet data can be difficult to interpret when it is spread across raw files or separate summaries. A structured analysis makes it easier to compare product categories and outlet segments, track headline metrics, and identify areas that may deserve further investigation.

Key analytical questions

- What are the total sales, average sales, item count, and average rating for the dataset?
- Which item types contribute the largest and smallest shares of sales?
- How does sales contribution vary between Low Fat and Regular product groups?
- How do sales differ by outlet size, outlet type, and location tier?
- How do outlet establishment-year cohorts compare in sales?
- Do the results remain consistent when the dataset is grouped and filtered in SQL, Python, and Power BI?

Scope of work

STAGE	ACTIVITIES	OUTPUT
Data understanding	Inspect columns, data types, row grain, categories, and business definitions.	Data dictionary and initial quality notes
Data preparation	Check missing values, duplicates, inconsistent labels, and invalid values.	Clean and consistent analysis dataset
Exploratory analysis	Summarise distributions, compare segments, and inspect outliers and trends.	EDA tables and observations
SQL investigation	Filter, join, group, aggregate, and reconcile business metrics.	Reusable queries and grouped summaries
BI reporting	Build measures, slicers, charts, and outlet comparisons.	Interactive Power BI report
Validation and communication	Cross-check calculations and write evidence-based findings.	Validated metrics and project summary

02 | Tools & Analytical Methods

TOOL	PURPOSE IN THE PROJECT
Python	Data profiling, missing-value and duplicate checks, type correction, exploratory data analysis, summary statistics, and validation. Common libraries: pandas, NumPy, and Matplotlib.
SQL	Querying and validating the data with filtering, joins, CASE expressions, GROUP BY, aggregate functions, and category/outlet-level summaries.
Power Query	Importing data and applying repeatable transformations such as type setting, column cleanup, null handling, and category standardisation.
Power BI	Communicating results with KPI cards, bar charts, donut charts, a trend visual, a comparison matrix, and interactive slicers.
DAX	Defining reusable measures for total sales, average sales, item count, average rating, and other metrics with clearly stated definitions.

Analysis workflow

1. Understand the dataset

Document what one row represents, what each field means, and which columns are suitable for grouping, filtering, or calculating metrics.

2. Prepare the data in Python

Profile nulls and duplicates, validate data types, standardise categories, inspect ranges, and run initial descriptive analysis.

3. Investigate in SQL

Build grouped summaries by item type, outlet type, outlet size, location tier, and establishment year; compare results with Python outputs.

4. Define metrics

Agree the business meaning and denominator for total sales, average sales, item count, rating, and item visibility.

5. Build the Power BI model

Load cleaned data, create relationships where needed, and use measures instead of hard-coded KPI values.

6. Visualise and test

Create report visuals and slicers; test cross-filtering, totals, labels, and edge cases against the validated analysis.

Expected deliverables

- A cleaned and documented dataset.
- Python notebook or script containing data-quality checks and EDA.
- SQL queries for KPI validation and segment-level analysis.
- Power BI report with measures, filters, and visual comparisons.
- A concise findings section supported by verified calculations.

03 | Metrics, Interpretation & Portfolio Summary

Metrics visible in the supplied report

TOTAL SALES	AVERAGE SALES	NUMBER OF ITEMS	AVERAGE RATING
$1.20M	$141	8,523	3.9

These values are transcribed from the screenshot. They should be treated as displayed report values until they are recalculated and reconciled against the source dataset.

How to interpret the analysis responsibly

- Use category and outlet comparisons to identify patterns, not to claim causes without additional evidence.
- Confirm whether Number of Items means distinct products, item records, or transaction rows.
- Define Average Sales precisely—for example, average per item, transaction, outlet, or row.
- Check the rating scale and whether a simple or weighted average is appropriate.
- Treat establishment year as an outlet cohort when that is what the underlying field represents.

Portfolio website description

Grokal Sales & Outlet Performance Analysis

An end-to-end retail data analytics project using Python, SQL, and Power BI to examine sales performance across product categories and outlet segments. The workflow covers data quality assessment, exploratory data analysis, SQL-based aggregation and validation, KPI definition, and interactive business reporting. The final Power BI report presents sales metrics and comparisons by item type, fat-content group, outlet size, outlet type, location tier, and establishment-year cohort, helping stakeholders explore performance and identify areas for further analysis.

Before publishing

- Only list Python and SQL as completed project tools if you actually used them in the analysis.
- Add two or three findings only after checking the source dataset and calculation logic.
- Verify currency, field names, aggregation rules, and filter behaviour before sharing the report publicly.